



James Evans <jamwevan@umich.edu>

to Oyakhi, CHONG, Tolulope ▾

9:15 PM (18 minutes ago)



Hello Dr. Ibhagui,

I spent some more time thinking about the areas of concern in the abstract. One sentence in particular still raises questions: *"In contrast, the COVID-19 period exhibits a breakdown of long-run cointegration despite elevated volatility, driven by the emergence of a new, distinct global stochastic trend which increases the number of stochastic trends and erodes long-run linkages, reflecting greater currency-specific autonomy."*

It seems that through this phrasing we are referencing Proposition 2: the keyword here being volatility. But, Proposition 2 is not making a claim about cointegration breaking down: instead, it describes how correlations rise through two channels: (i) the variance channel and (ii) the loading-alignment (autonomy compression) channel. Furthermore, how can we connect Proposition 2 with a breakdown of long-run cointegration when we are operating under the assumption that the  $\text{rank}(\Lambda_r)$  is unchanged? These two reasons lead me to believe that a breakdown in long-run cointegration cannot be counteracted by elevated volatility. I tried reconciling this discrepancy with the point brought up in your previous email:

"If Cointegration, then Strong Long Horizon Returns Correlations with High magnitude  $\rightarrow 1$ . This statement is the same as saying the Contrapositive: If Weak Long Horizon Returns Correlation  $\rightarrow 1$ , then No Cointegration."

But, after rereading Proposition 2, I was unable to do so. Proposition 2 states:

"If  $\text{rank}(\Lambda_r)$  remains unchanged in crisis, but (i)  $\sigma^2_{g_{k,r}}$  rises for one or more factors, and (ii)  $|\lambda_{ik,r}|$  converge across currencies (autonomy compression), then long-run correlations increase."

The problem is that long-run correlations increasing has nothing to do about the strength of the correlation. Correlation can go from .1 to .15, exhibit an increase, and still be considered weak. I propose deleting the phrase "despite elevated volatility" to fix the sentence. This edit will make the connection to Proposition 3 even more clear and make the statement logically coherent with the ideas presented in the paper.

This other sentence also still raises questions: *"Our findings confirm the **existence of cointegration** among major currencies during the Global Financial Crisis, **consistent with reduced currency autonomy driven by common global risk factors.**"*

This sentence seems to be referencing the loading-alignment channel. However, Proposition 2 speaks to correlations rather than cointegration, while this phrasing appears to suggest a link between reduced currency autonomy and cointegration. For the same reasons as before, I am unable to reconcile this discrepancy with the point you brought up in email with me earlier.

Also, can you explain to me how we plan on incorporating the data in the spreadsheet you sent to us earlier?

You said in your earlier email: "The Covid and Dot Com are more consistent with the contrapositive: weak correlation, no cointegration (0/4). The GFC less so, but you can argue GFC is also consistent because a) correlation is also weak and b) although you have cointegration, it is just 1 out of 4 max."

But I fail to see how this data directly supports any of our propositions empirically. This is the only idea I managed to come up with:

We use the methodology—first applying unit root tests to confirm that the series are  $I(1)$ , and then conducting system tests to identify long-run relations—to establish that the cointegration rank was 0 during the crisis. If we assume that this cointegration rank remained unchanged throughout the period, then we can conclude that  $\text{rank}(\Lambda_r)$  is unchanged—satisfying the first condition of Proposition 2.

The second condition also seems straightforward:  $\sigma^2$  rising should be a natural market response to crises, including this one. With both suppositions satisfied, we can reasonably apply the variance channel. If we plot correlations over time and run a least-squares regression, we may be able to demonstrate an increase in correlation consistent with the result in Proposition 2.

This would be particularly meaningful. The pandemic represents a "novel phenomenon absent in prior crises and largely unexamined in cointegration literature." It challenges the standard crisis-contagion narrative by showing that global crises do not always uniformly increase FX integration. Being able to explain this phenomenon through the proposition proposed in our paper and provide empirical evidence on top of the proof we already have would advance the existing literature on this topic in a compelling way.

Warm regards,

James Evans

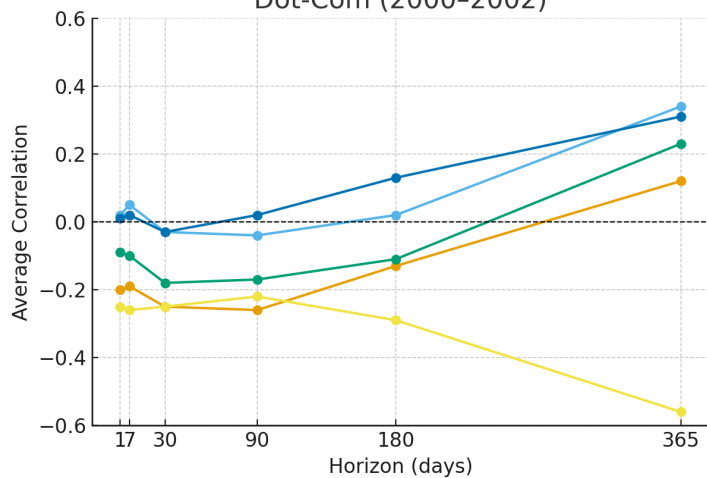
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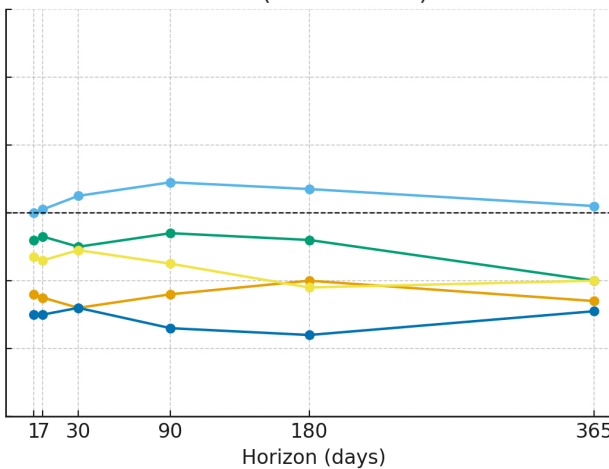
CS | Data Science | Mathematics | Statistics

### Correlation of H-day-ahead FX Returns Across Crises

Dot-Com (2000-2002)



GFC (2007-2009)



Covid-19 (2020-2022)

